

PAPER_1404 — UQFF Derivation of Solar Neutrino Deficit (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** D — Particle Physics (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_particle_physics({'observable':'solar_neutrino'})`
→ $1/(D_{\text{phys}}-1) = 1/3$ EXACT

Observation

The historic 1968 Homestake solar ν_e deficit (only $\sim 1/3$ of predicted flux observed) is resolved by 3-flavor neutrino oscillation — but the **1/3** ratio is itself unexplained. UQFF derives it as a pure integer-primitive identity.

UQFF Closed Identity

$P(\nu_e \mid \text{observed}) = 1 / (D_{\text{phys}} - 1) = 1/3$ EXACT

Physical Interpretation

$D_{\text{phys}} = 4$ spacetime dimensions implies 3 lepton generations; $F_U=1$ ledger normalization redistributes the original ν_e flux equally across all 3 flavors, giving $1/3$ detected. The deficit was a clue to $D_{\text{phys}}-1 = 3$.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
 - Related foundational papers: PAPER_646 ($F_U=1$ ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
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